Chapter 13. Indicator Variables

Dummy variables, interactions, several categories, piecewise linear regression

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13.1 One indicator variable

Qualitative explanatory variables (sex, treatment, region) enter a regression through **indicator** (dummy) variables taking the values 0 and 1.

Parallel lines. Suppose y is the score on a test, x_1 the hours of preparation, and $x_2 = 1$ for students who attended a review session and 0 otherwise. The model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2),$$

gives two response functions with a common slope:

$$E(y) = \begin{cases} \beta_0 + \beta_1 x_1, & x_2 = 0, \\ (\beta_0 + \beta_2) + \beta_1 x_1, & x_2 = 1. \end{cases}$$

β_1 is the effect of x_1 in both groups, and β_2 is the vertical distance between the parallel lines: the group effect at any fixed x_1. Fitting one model, rather than one line per group, uses all n observations to estimate the common slope and σ^2.

Interaction. If the effect of x_1 may differ between groups, add the product term:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i1} x_{i2} + \epsilon_i, \qquad E(y) = \begin{cases} \beta_0 + \beta_1 x_1, & x_2 = 0, \\ (\beta_0 + \beta_2) + (\beta_1 + \beta_3) x_1, & x_2 = 1. \end{cases}$$

Now β_2 is the difference in intercepts and β_3 the difference in slopes. The hypotheses of Chapter 5 become ordinary tests in one model:

- parallel lines: $H_0 : \beta_3 = 0$ (a t test);
- identical lines: $H_0 : \beta_2 = \beta_3 = 0$ (a partial F test with 2 degrees of freedom).

These agree with the separate-fit tests of Section 5.3 because the interaction model with $n_1 + n_2$ observations has exactly the four parameters of the two separate lines and the same residual sum of squares.

Example 13.1

In a study of 20 firms, y is the time to adopt an innovation, x_1 the firm size and $x_2 = 1$ for stock companies. The fit $\hat{y} = 33.87 - 0.102 x_1 + 8.06 x_2$ says that adoption is faster for larger firms (about 1 month per 10 size units) and that stock companies take about 8 months longer at every size. Adding the interaction gives $\hat{\beta}_3 = -0.0004$ with standard error 0.018, so $t_0 = -0.02$: the parallel-lines model is adequate.

13.2 Several categories

A qualitative variable with q categories needs $q - 1$ indicators. With three categories E_1, E_2, E_3, define

$$x_2 = \mathbf{1}\{E_1\}, \qquad x_3 = \mathbf{1}\{E_2\},$$

with E_3 the **baseline**:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \epsilon_i, \qquad E(y) = \begin{cases} (\beta_0 + \beta_2) + \beta_1 x_1, & E_1, \\ (\beta_0 + \beta_3) + \beta_1 x_1, & E_2, \\ \beta_0 + \beta_1 x_1, & E_3. \end{cases}$$

So β_2 and β_3 compare E_1 and E_2 with the baseline, and $\beta_2 - \beta_3$ compares E_1 with E_2.

The dummy variable trap. Using q indicators together with an intercept makes the columns of X linearly dependent: the indicators sum to the intercept column $\mathbf{1}$. Then $\operatorname{rank} X <$ number of columns, $X'X$ is singular, and β is not identifiable. Either drop one indicator (baseline coding, as above), drop the intercept, or impose a constraint such as $\sum(\text{effects}) = 0$.

Interactions and several factors. Separate slopes are obtained with the products x_1x_2 and x_1x_3:

$$E(y) = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_3 + \beta_4x_1x_2 + \beta_5x_1x_3,$$

so that in E_1 the line is $(\beta_0+\beta_2)+(\beta_1+\beta_4)x_1$, and so on. Several qualitative factors each get their own set of indicators, and interactions between factors are products of their indicators. A model containing only indicators (no quantitative regressors) is an analysis of variance model written as a regression.

13.3 Piecewise linear regression

Sometimes the slope changes at a known point x_w (a threshold or a policy change), while the response stays continuous. With $x_2 = \mathbf{1}\{x_1 > x_w\}$,

$$y_i = \beta_0 + \beta_1x_{i1} + \beta_2(x_{i1} - x_w)x_{i2} + \epsilon_i,$$

so that

$$E(y) = \begin{cases} \beta_0 + \beta_1x_1, & x_1 \le x_w, \\ (\beta_0 - \beta_2x_w) + (\beta_1 + \beta_2)x_1, & x_1 > x_w. \end{cases}$$

The two lines meet at x_w, β_2 is the change in slope, and $H_0 : \beta_2 = 0$ tests for a break. The regressor $(x_1 - x_w)_+ = (x_1 - x_w)x_2$ is a **truncated power basis** function; several knots $x_{w_1} < x_{w_2} < \cdots$ add one such term each, giving a linear spline. If a jump at x_w is also allowed, add x_2 itself.

Example 13.2

Unit cost y against lot size x_1 for eight lots, with a suspected change at $x_w = 500$:

x_1	300	340	400	480	570	650	720	800
y	475	440	452	377	355	257	249	139

The fit is $\hat{y} = 589.5 - 0.395x_1 - 0.389(x_1 - 500)_+$: the cost falls by 0.40 per unit up to 500 and by $0.40 + 0.39 = 0.78$ per unit beyond.